RAAY — 1-PAGE SOLUTION PITCH

Sovereign Indian Voice Operating System for Windows 11 | Powered by Gnani.ai (Prisma STT & Timbre TTS)

GitHub: https://github.com/vakrahul/Raay | Team: Rahul Vakiti & Samyutha | #GreatIndianAIInternshipChallenge #GnaniAI

1. THE PROBLEM: THE ENGLISH TYPING BARRIER & BROWSER-TRAPPED AI

Over a billion Indians think and speak fastest in Telugu, Hindi, Tamil, Kannada, Malayalam, Bengali, or code-mixed Hinglish/Telugish — yet operating a computer still requires typing complex prompts in English. Meanwhile, standard AI chatbots are trapped inside a browser tab: they only talk back in text and score 0/5 on real local desktop execution.

2. OUR SOLUTION: RAAY (ZERO-TYPING SOVEREIGN VOICE OS)

Raay turns Gnani.ai sovereign speech models into an always-on, hands-free Windows 11 Voice Operating System. You speak naturally in your own Indian language from across the room — Raay plans, executes, and verifies multi-step workflows directly across Microsoft Excel, MS Paint, VS Code, Chrome, Notepad, and Windows File Explorer.



3. GNANI.AI SOVEREIGN STACK & PLUGGABLE LLM ADAPTERS USED

- **Gnani Prisma STT (WSS /stt/v3/stream & POST /stt/v3):**
  Streams 16kHz PCM16 frames with server VAD (280ms silence), telephonic noise cancellation (nc=true), keyword biasing, and code-mixed recognition across 10+ Indian languages.
- **Gnani Timbre v3.0 & v2.5 TTS (POST /api/v1/tts/sse):**
  Streams context-aware Indic speech with authentic Rupee/Lakh/Crore cadence, automatic RIFF header normalization, and a <1ms SHA-256 disk cache.
- **Pluggable External LLM Adapter Layer (src/providers/):**
  Connects Anthropic Claude, OpenAI GPT-4o, Google Gemini Flash/Pro, or local Ollama before decomposing spoken requests into deterministic OS automation tasks.

4. FIVE LIVE SEEN DESKTOP BENCHMARKS (RAAY: 5/5 vs. CHATBOT: 0/5)

1. Live Kirana/Grocery Excel BI & GST Audit:
Opens 25-item Excel sheet, builds live COM Bar/Pie Charts, and speaks exact UPI (Rs.2,478), Card (Rs.1,323), Cash (Rs.940) & GST (Rs.249) totals.

2. Multi-Object MS Paint Creation via UIA:
Opens MS Paint, inspects UIA ribbon controls, selects shapes & colour swatches, and draws pixel-verified scenes live on screen.

3. Sub-200ms Voice Barge-In & Recolour:
Interrupts active drawing mid-stroke on spoken Telugu/Hinglish command, triggers Ctrl+Z undo, and recolours the target shape in-place.

4. Cross-App Design-to-Code Export:
Exports the active MS Paint SceneSpec into a responsive HTML5/SVG + JSON blueprint and opens the project directly in VS Code.

5. Local Downloads Report Summarizer:
Scans ~/Downloads for the latest report, extracts key points into a bilingual summary, and opens it automatically in Notepad.

5. TEAM CONTRIBUTIONS & REAL-WORLD IMPACT

- Rahul Vakiti: Systems Architecture, Gnani Prisma STT & Timbre v3.0 Streaming Pipeline, Windows 11 UIA / Excel COM / MS Paint Execution Controllers, Sub-200ms Barge-In Engine, and Windows DPAPI Hardware Security Core.
- Samyutha: Multilingual Conversational Planner (te-IN, hi-IN, ta-IN, hi-en, en-IN), Indic Script Ratio Verification, Kirana/Grocery BI Dataset & Analytics Modelling, and 3D Volumetric Web Cockpit UX.